

The Hong Kong Polytechnic University

Department of Land Surveying and Geo-Informatics

LSGI3102 Land Surveyors in Society

Project 2: Case Study on MTR Shatin-Central Link Construction Scandal

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Report Written: 24th November [REDACTED]

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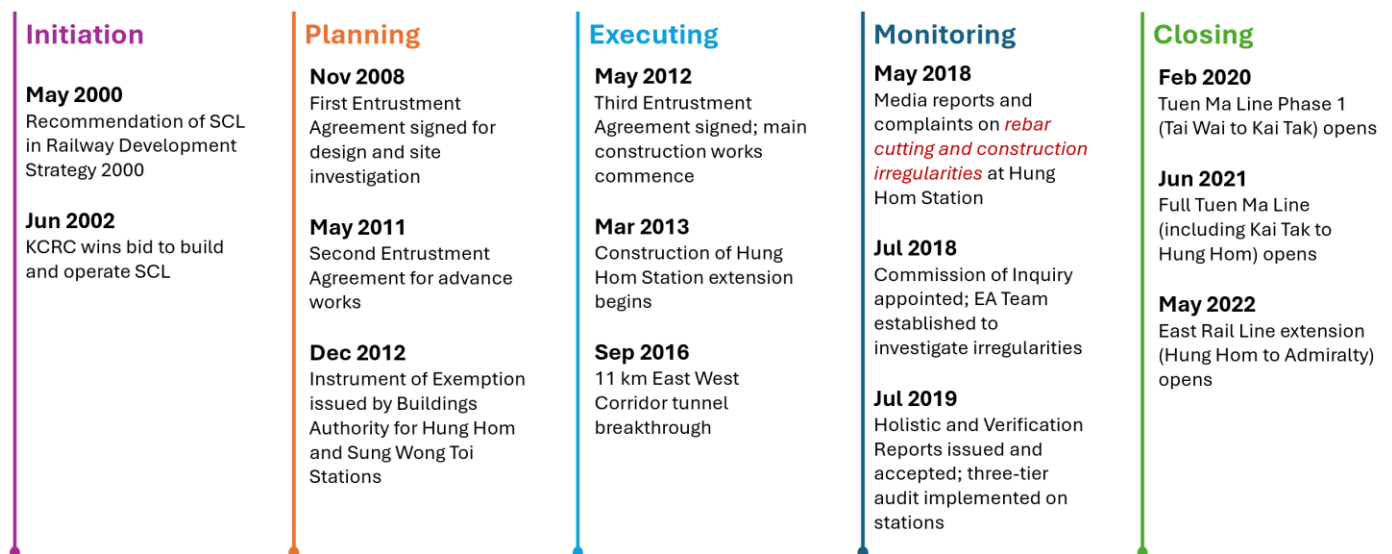
I. Overview of the MTR Shatin-Central Link Project

Mass transit is essential for daily life in Hong Kong. Originating from the Railway Development Strategy 2000, the Shatin to Central Link (“SCL”) project was proposed to expand railway coverage, connecting north-south and east-west regions through strategic corridors. Its key aims include improving connectivity and alleviating severe road traffic congestion in East Kowloon. As a large-scale infrastructure spanning 17 km with 10 stations, it involved extensive community engagement, costs exceeding HKD 90 billion, and a timeline from 2012 to 2022 (HKIE, 2020). Ensuring quality, safety, and resource balance is vital. However, scandals since 2018 have exposed issues. This case study focuses on conformance to rules of conduct and local regulations throughout the SCL project.

Extracted from: (1) Railway Development Strategy 2000; (2) The Shatin to Central Link Project Interim Report No. 1; (3) The Shatin to Central Link Project Final Report (Volume 1 & 2)



MTR Shatin-Central Link Project Key Events and Documentations Overview



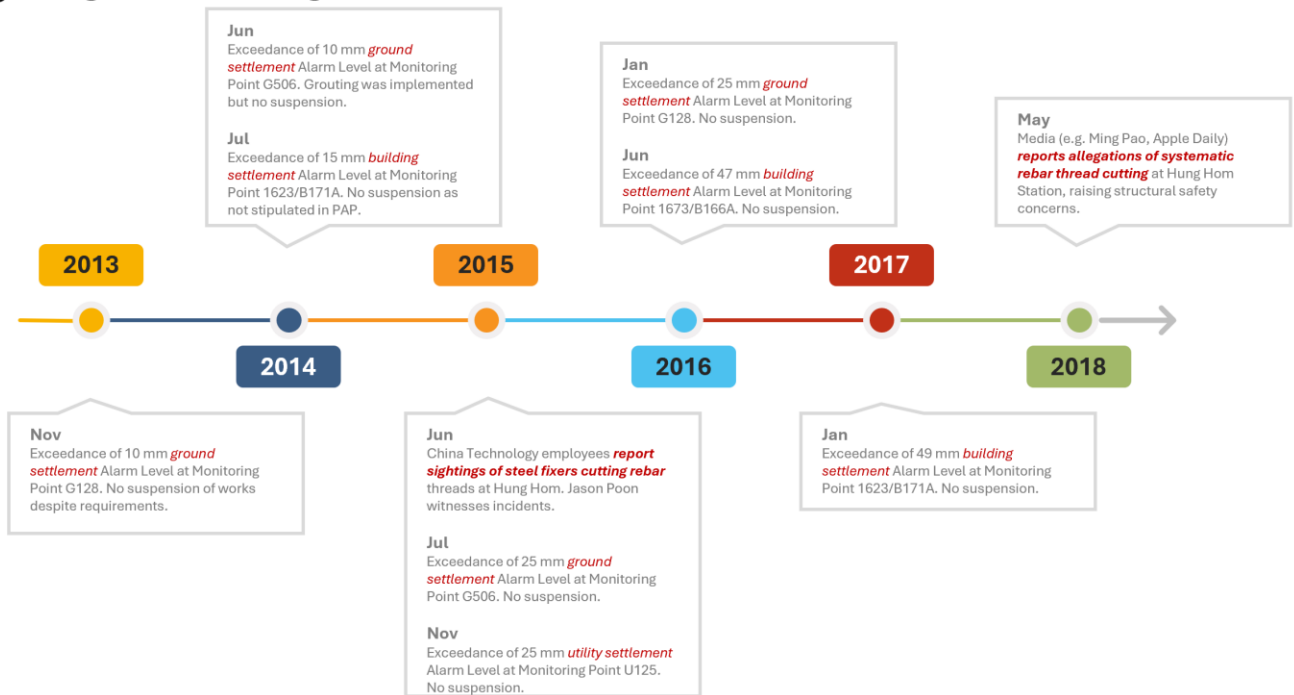
II. Timeline of Events

During the SCL construction phase of 2013 to 2018, irregularities included multiple exceedances of Alert, Alarm, and Action ("AAA") levels in ground, building, and utility settlements without work suspensions or root-cause investigations. Worker reports of illicit rebar cutting in 2015 and 2017 led only to internal probes by the main contractor, resulting in warnings without public disclosure. Media exposure in 2018 triggered government involvement, appointing COI for hearings and evidence collection. Findings uncovered unauthorized coupler changes and unapproved structural revisions. Leighton pleaded guilty and was fined in 2023.



MTR Shatin-Central Link Project Irregularities During Construction Phase

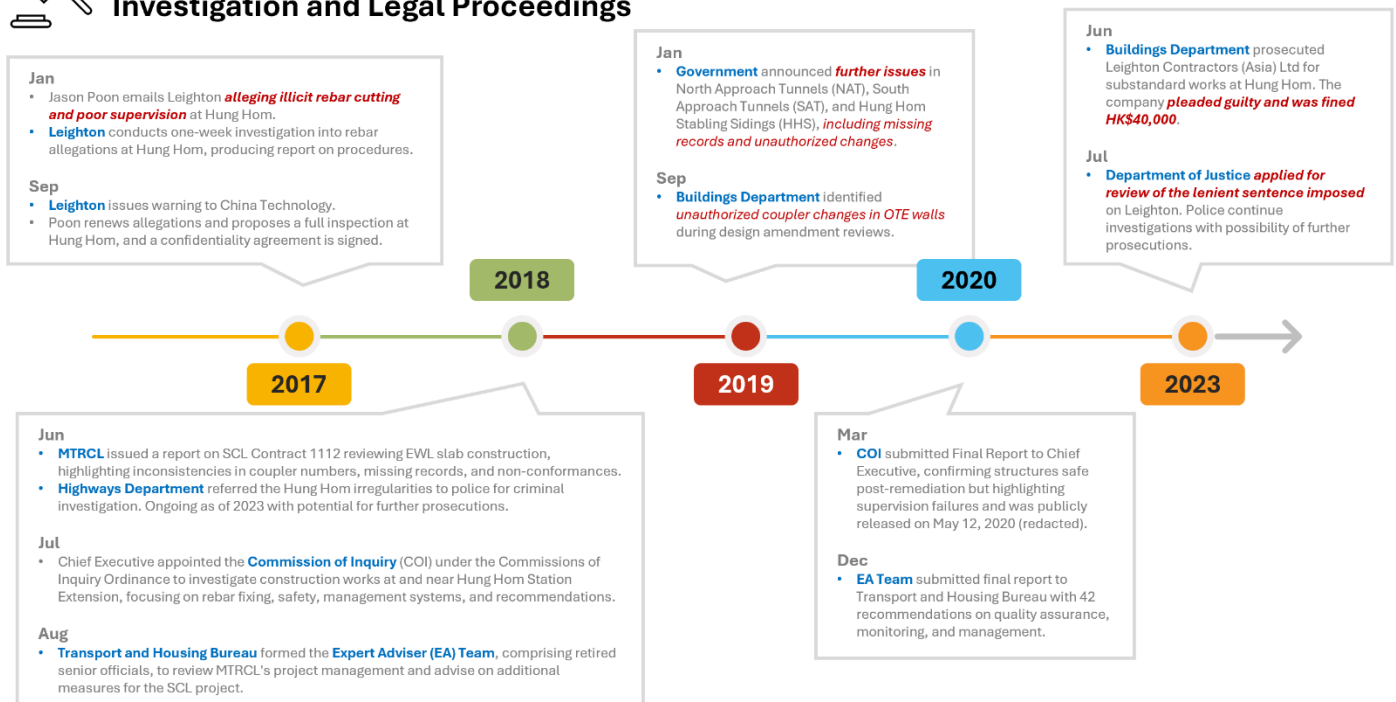
Extracted from: Final Report of Commission of Inquiry into the Construction Works at and near the Hung Hom Station Extension under the Shatin to Central Link Project



Extracted from: (1) Final Report of Commission of Inquiry into the Construction Works at and near the Hung Hom Station Extension under the Shatin to Central Link Project; (2) The Shatin to Central Link Project Final Report (Volume 1 & 2)

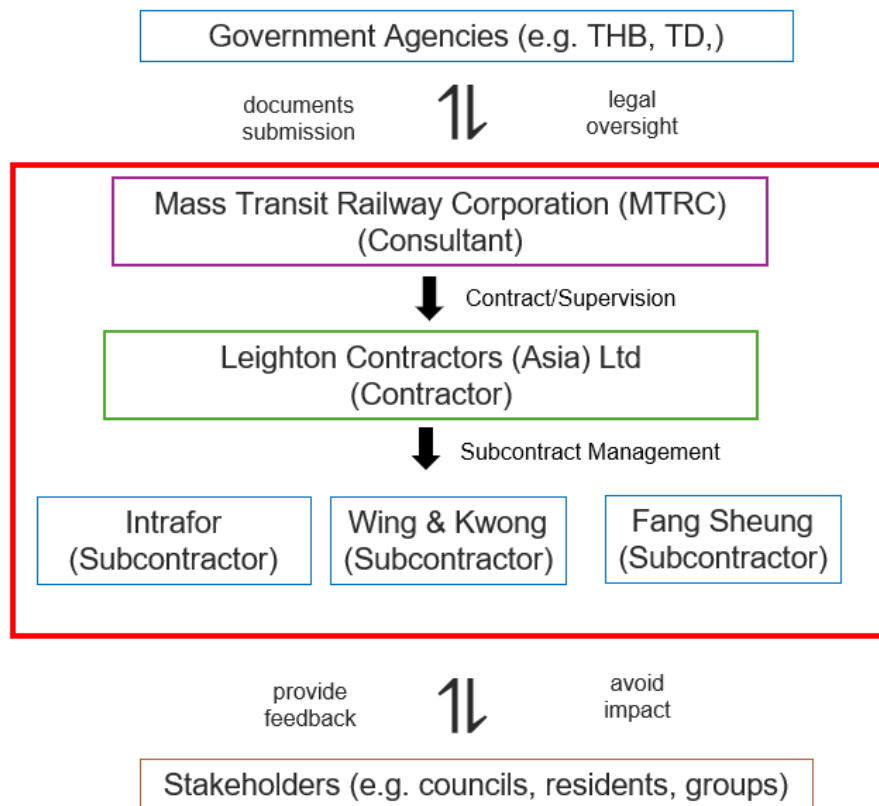


MTR Shatin-Central Link Project Investigation and Legal Proceedings



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III. Key Stakeholders and Their Roles



The SCL project involves multiple key stakeholders with distinct roles and responsibilities. The MTR Corporation (“MTRC”) is the project owner and the main consultant. This body was mandated to undertake project planning and design supervision, works supervision, assuring the whole process works within the defined standards through their reports presented to the government in Hong Kong. Notably, the corporation was also expected to ensure the process adheres to all applicable safety and construction standards.

The main contractor, Leighton Contractors (Asia) Limited, is responsible for executing the construction works in accordance with the approved design, contract specifications, and statutory requirements. Its duties include proper site management, workmanship, documentation, and coordination of subcontractors.

Subcontractors such as Intrafor, Wing & Kwong Steel Engineering Co. Ltd, and Fang Sheung Construction Company are responsible for carrying out specialized construction tasks, including bar fixing, coupler installation, structural steelwork and temporary works. They must follow approved design drawings, method statements and quality control procedures, and are required to provide accurate site records, test reports and as-built documentation to the main contractor and the client.

IV. Previous Criminal and Ethical Records

Mass Transit Railway Corporation

In earlier projects, MTRC had already been criticized for serious governance and transparency problems. During the Guangzhou–Shenzhen–Hong Kong Express Rail Link project, an independent expert panel led by Mr Justice Michael Hartmann found that MTRC failed to report cost overruns and delays to the Transport and Housing Bureau, and that its poor internal communication left senior managers and the government unaware of the true situation (Cheng, 2018). These findings suggested weaknesses in MTRC’s project control and accountability culture even before the Sha Tin–Central Link scandal.

Leighton Contractors (Asia) Ltd

In Hong Kong, Leighton Contractors (Asia) Ltd has already been convicted in relation to the Sha Tin–Central Link works. The Kowloon City Magistrates’ Courts fined the company HK\$40,000 after it admitted failing to follow an approved construction plan when building a 131.4-square-metre air-conditioning room at Hung Hom Station in 2018 (The Standard, 2023). This conviction illustrates non-compliance with statutory and design requirements and reinforces broader ethical concerns about Leighton’s site management and quality-control practices.

V. Possible Professional Misconduct

A. Misconduct During Project Execution

The scandal first emerged in May 2018, it was discovered that workers at the Hung Hom station expansion had cut short steel bars instead of screwing them properly into couplers (Ho, 2020). This fundamentally compromised the structural integrity that millions of workers rely on every day. What makes it even more troubling is that the Leighton Asia instructed the workers to shorten the steel bars, so that it could fit into defective couplers, effectively faking a secure connection (Circle19, 2018). This wasn't a mistake, but a deliberate concealment intended to hide problems rather than fix them properly.

Even more upsetting is that a subcontractor reported this case to Leighton, even with photographic evidence of workers installing shortened steel bars in Sep 2015, yet the illegal operations continued (Circle19, 2018). These actions reveal a culture where it was apparently acceptable to cut corners. Those leads to professional misconduct as it relates to honesty and integrity.

B. Misconduct in Legal Processes

The misconduct was not limited to the construction site, but also extended to the handling of evidence and time records during the investigation. Commission chairman Michael Hartmann described MTR Corp's practice of backdating records as "a worrying feature" that undermined the credibility of its records (Young Post Club, 2018). What this mean is that, rather than being forthcoming upon detection of an issue, key documentation was manipulated to misrepresent that due diligence had been consistently applied all along.

The most shocking thing is when the MTRC investigated after the incidents emerged in early 2019, the contractor could not provide a large number of its as-built records (Zhang, 2020). These were not just administrative forms, also with the documents providing like proving the work had been done correctly and safely. CEO of MTRC Lincoln Leong also acknowledged that corruption allegations against Leighton were reported to the ICAC but deliberately omitted from public reports due to defamation concerns (Young Post Club, 2018). These actions of concealment expose a worrying lack of transparency and accountability in legal procedures. It seriously undermining public trust in the profession.

need to state which rules of which professional bodies (e.g. HKIE, HKIS, etc.)

VI. Violations of Specific Codes of Practice, Regulations, and Ordinances

A. Statutory Violations under Hong Kong Ordinances

This case appears to violate Cap. 123 Building Ordinance in Section 40 (2A) for deviating from material use of works in the approved plans. For example, defective couplers were used, which likely violates subsection (a)(i) “are defective or do not comply with the provisions of this Ordinance”. In (a)(ii) forbids any material handling that deviates from the ordinance. In (b) forbids materials that deviate from works in the plan approved by the Building Authority. Overall, Leighton likely violated Cap. 123.

This case also appears to violate Cap. 509 Occupational Safety and Health Ordinance for imposing a working risk or hazard to the workers. In this case, the steel reinforcement has not been tightened to the concrete slabs, and underground water seepage problem was severe. This structural fault might have caused concrete spalling or even structural collapse, posing a danger to the workers on the site. Section 6 states that employers must ensure the safety and health of workers. Since water seepage and potential spalling risks were apparent, a safe system of work was not established, and thus likely violates Cap. 509.

B. Professional Misconducts

This case appears to violate RICS rules of conduct Rule 1.5. Without gaining the approval and deviating from the approved plans, it can be considered dishonest behavior under the RICS rules of conduct.

1.5 Members and firms providing advice and opinion in a professional context do so honestly and objectively based on relevant and reliable evidence, and firms have processes to ensure that directors, partners and employees do so.

This case appears to violate Rule 2.2. It did not appear that the workers were well-informed and supervised to correctly install the steel reinforcement to the walls or to avoid the use of faulty materials. This shows that Leighton’s supervision was inadequate to ensure workers had the necessary knowledge for the project, thus likely violating Rules 1.5 and 2.2 under the RICS rules of conduct.

2.2 Members and firms supervise any employees undertaking work for them and ensure that these employees have the necessary knowledge, skills and resources to do their tasks competently.

This case also appears to violate HKIS rules of conduct. The engineering works were shown to deviate from the approved plans, which suggests supervisors and inspectors had incompetence in professional standards, and violated Rule 1.1.3.

1.1.3 not accept any instruction which he is not qualified or competent to discharge;

If they had knowingly continued the faulty engineering works, then they would have likely violated Rule 1.1.4, since they were being dishonest to the Building Authority.

1.1.4 not give opinions in his professional capacity that are not, to the best of his ability, objective, reliable and honest;

In a nutshell, this shows that the key personnel, such as engineers and supervisors in Leighton, might have violated HKIS rules of conduct.

VII. Impact Summary of the Scandal

A. Project delays & Safety issues

The scandal of the Shatin-Central Link caused severe delays. Based on the Final report issued by the Transport and Housing Bureau in December 2020, the construction defects discovered at Hung Ham Station compelled the large-scale, extensive remedial works and comprehensive structural assessment. In order to investigate the stability of thousands of steel bars, review the structural integrity and implement remedial measures (Lau et al., 2020a). This led to the project timeline being significantly delayed, far exceeding the original deadline and costing an extra \$2 billion to reinforce the structure. Although these investigative processes are necessary for public safety, the construction schedules were severely postponed. Hence, the water seepage problem aggravates the safety risks of the station, combining with the possibility of concrete spalling, corrosion of steel and may further lead to structural deterioration over time. This could cause potential damage to the future user.

B. Reputation Damage

This scandal may cause serious damage to the MTR, the related government department and the engineering firm. As a public services provider, the MTR may lose its trust in the public, questioning its capability in managing and monitoring large-scale infrastructure projects. Hence, the engineering firm also suffered reputation damage in the construction project. Although the company admitted its guilt and was fined HK\$40000, this verdict was viewed as lenient (Wong, 2023). Some people accuse the government of imposing leniency to shield the MTR Corporation, raising public suspicion of systemic corruption and privileged treatment. This controversy drew media and legislative council attention, the legislative councillor questioned why more serious charges were not being pursued, giving evidence of widespread construction defects in thousands of steel. This reputation damage undermined the public confidence in the regulatory mechanism and quality assurance system of Hong Kong's construction industry.

C. Possible Casualties

Although there are no actual casualties were reported, the potential for serious harm was significant and alarming. The commission's investigation reveals that the scale of improper work was wide-ranging, affecting numerous stations and greatly increasing the chance of structural failure. This may lead to platform collapse, falling ceiling or structural failure that could have trapped or injured large numbers of people in an underground environment.

VIII. Court Judgment and Decisions

A. Facts / Case Background

On June 30, 2017, MTR Corporation engineers inspected the construction status of the reinforcement works at Hung Hom MTR station and found that the steel bars on the floor slab of a variable refrigerant volume room were not fully fixed to the connectors (Wong, 2023). Although Leighton had been instructed by the railroad operator to rectify the problem before pouring concrete, its workers continued pouring that day (Wong, 2023). In mid-2018, a government inquiry was launched after it was discovered that the steel bars had been cut during installation, causing the project to be delayed by more than a year (SCMP Editorial, 2023). In May 2019, the violation was exposed when an MTR Corp reported the construction violation to the Building Authority (Wong, 2023).

B. Issues

Whether Leighton Contractors (Asia) Limited violated the Buildings Ordinance by deviating from plans approved by the Building Authority (Wong, 2023).

C. Rules and Reasoning

Leighton was charged under Section 40(2A)(b) of the Buildings Ordinance for “deviated in a material way from the work shown in plans approved by the Building Authority”, with a maximum fine of HK\$1 million (Wong, 2023).

D. Application and Analysis

On June 27, 2023, Leighton pleaded guilty to violating the Buildings Ordinance by failing to comply with approved plans when constructing a 1,410-square-foot space at Hung Hom MTR station. (Wong, 2023). Leighton's lawyers argued in court that it was a minor "technical" error and pointed out that this was Leighton's first offence (SCMP Editorial, 2023). Magistrate Kestrel Lam Tsz-hong agreed that the quality issue involved a limited area, caused no injuries, and had no further structural defects (Wong, 2023).

E. Conclusion

On June 27, 2023, the Kowloon City Court imposed a fine of HK\$40,000 on Leighton Contractors (Asia). The fine was reduced from HK\$60,000 to HK\$40,000 because of the company's guilty plea (Wong, 2023). The Buildings Department referred the case to the Contractors' Disciplinary Board for instigating disciplinary proceedings under the Buildings Ordinance (Buildings Department, 2023)

IX. Lessons Learned

Through this lesson, we have gained valuable insights into the importance of professional legal knowledge and how it can be leveraged to combat illegal activities within the field of land surveying. In contemporary society, numerous engineering projects are underway, and it is not uncommon for land surveyors to encounter unlawful behaviors among clients or other personnel. For instance, the MTR Shatin-Central Link incident represents only a small fraction of the legal and ethical challenges faced in our profession.

Therefore, Land surveyors should always strictly follow legal regulations to avoid potential casualties and other accidents or mishaps. Since we know how important our job is, we should always be watchful and proactive. If illegal activities are found, it is our duty to inform the right agencies about these problems. In so doing, not only is the protection of the public ensured, but also the integrity of our job and the confidence invested in us by society.

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